PERFORMANCE TESTING REPORT

Project: ComicCraft – AI Comic Story Creator using Gemini Models

Team ID	SWTID-2026-3296	**Team Size**	4
Team Leader	Kamalesh P R	**Test Date**	15 March 2026 (template date)
Team Members	JAYAKUMARAN V M Kannan P Naresh C	**Maximum Marks**	5 Marks

Important note: The supplied project brief describes the application and its endpoints, but does not include executed performance-test measurements or tool screenshots. Accordingly, actual values are marked “Not measured” rather than invented. Run the tests and replace these fields with observed results before submission.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter / Locust / k6 (select the tool actually used; not specified in the supplied project details)
Type of Testing	Load testing and stress testing (planned test types)
Target Module / API	Comic generation workflow: POST /generate; JSON API: /generate-comic/json; image test: /test-image
Test Environment	Local development server using FastAPI + Uvicorn (uvicorn app.main:app --reload), as described in the project brief
Test Date	15 March 2026 (date shown in supplied template; update to actual execution date if different)

Step 2: Test Scenarios

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**S.No**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='S.No', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Test Scenario / Description**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Test Scenario / Description', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Virtual Users**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Virtual Users', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Duration (sec)**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Duration (sec)', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Expected Outcome**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Expected Outcome', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
1	Baseline load on homepage and API documentation	1	60	Homepage and /docs respond without errors.
2	Concurrent requests to /generate-comic/json with valid prompts	5	180	Requests are processed; response time and error rate remain within defined targets where feasible.
3	Concurrent comic generation through /generate form route	10	300	System handles concurrent submissions without unexpected server failures.
4	Stress test by increasing concurrent requests to comic-generation endpoints	20	300	Identify saturation point; record latency, errors, CPU and memory. Degradation is documented.

Step 3: Performance Test Results

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**S.No**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='S.No', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Metric**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Metric', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Target Value**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Target Value', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Actual Value**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Actual Value', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Status**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Status', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**Remarks**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Remarks', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
1	Average response time	< 2 seconds	Not measured	Pending	Enter measured average from test report.
2	Maximum response time	< 5 seconds	Not measured	Pending	Enter maximum observed latency.
3	Throughput (requests/sec)	Record achieved throughput	Not measured	Pending	Record completed requests per second.
4	Error rate	< 1%	Not measured	Pending	Record failed requests as a percentage.
5	CPU utilization	< 80%	Not measured	Pending	Capture peak/average CPU during test.
6	Memory utilization	< 80%	Not measured	Pending	Capture peak/average memory during test.

Step 4: Observations & Analysis

No measured performance observations were included in the supplied materials. Execute the scenarios above and document response-time trends, throughput, errors, and resource utilization here. Comic generation invokes external Gemini and Stable Diffusion services, so record whether elapsed time includes external model/API processing and note any API rate limits or network delays. Do not mark a metric Pass or Fail until actual measurements are available.

Observation notes (to complete after execution):

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Step 5: Screenshots / Evidence

Attach the actual JMeter/Locust/k6 report or screenshots, including the run configuration and summary graphs. Evidence was not present in the supplied files and is therefore not fabricated in this report.

Evidence checklist	Attach / complete
Test tool dashboard or summary report	Pending attachment
Latency / throughput graph	Pending attachment
CPU and memory utilization evidence	Pending attachment
Execution date and environment confirmation	Pending confirmation